

Chapter 22: Change of basis

Recall: If V be a vector space of dimension n , with ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ for V , then we can build a coordinate representation of $\mathbf{v}$:

$$C_B(\mathbf{v}) = C_B(a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \dots + a_n\mathbf{b}_n) = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \dots + a_n\mathbf{e}_n = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Example: Consider P_1 with basis $B = (1, x)$. Find $C_B(p)$ for $p = x - 10$.

$$C_B(p) = \begin{bmatrix} -10 \\ 1 \end{bmatrix}$$

Consider P_1 with basis $E = (2x + 1, 3x - 2)$. Find $C_E(p)$ for $p = x - 10$.

$$\begin{aligned} p = x - 10 &= a_1 \cdot (2x + 1) + a_2 \cdot (3x - 2) \\ &= (2a_1 + 3a_2) \cdot x + (a_1 - 2a_2) \\ \Rightarrow \begin{cases} a_1 = -4 \\ a_2 = 3 \end{cases} &\Rightarrow C_E(p) = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \end{aligned}$$

What if there was a way to easily move from one basis to another?

Question: Suppose V is 3-dimensional, with $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ an ordered basis for V .

Suppose $\mathbf{v}$ and $\mathbf{w}$ are two vectors in V with $C_B(\mathbf{v}) = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ and $C_B(\mathbf{w}) = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

What is $C_B(\mathbf{v} + \mathbf{w})$?

$$C_B(\mathbf{v} + \mathbf{w}) = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix} = C_B(\mathbf{v}) + C_B(\mathbf{w})$$

What is $C_B(k\mathbf{v})$ for a scalar k ?

$$C_B(k \cdot \mathbf{v}) = k \cdot C_B(\mathbf{v})$$

Fact: For a vector space V with ordered basis $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$,

- $C_B(\mathbf{v} + \mathbf{w}) = C_B(\mathbf{v}) + C_B(\mathbf{w})$ for $\mathbf{v}, \mathbf{w}$ in V
- $C_B(k\mathbf{v}) = kC_B(\mathbf{v})$ for $\mathbf{v}$ in V and k in $\mathbb{R}$.

Definition: Let $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and $D = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be two ordered bases of an n -dimensional real vector space V .

Suppose $\mathbf{v}$ is a vector in V . We could use either basis B or D to write out a coordinate representation for $\mathbf{v}$:

$$C_B(\mathbf{v}) = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \text{ means } \mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n$$

$$\text{and } C_D(\mathbf{v}) = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \text{ means } \mathbf{v} = c_1 \mathbf{w}_1 + \dots + c_n \mathbf{w}_n$$

Goal of this section: Build a matrix P so that $PC_B(\mathbf{v}) = C_D(\mathbf{v})$ for all $\mathbf{v}$.

Let's view P as $P = [\mathbf{p}_1 \ \mathbf{p}_2 \ \dots \ \mathbf{p}_n]$ (i.e. the i th column of P is $\mathbf{p}_i$).

What is $C_B(\mathbf{v}_1)$?

$$C_B(\mathbf{v}_1) = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

If we want $PC_B(\mathbf{v}_1) = C_D(\mathbf{v}_1)$, what must be true about P ?

$$P \cdot C_B(\mathbf{v}_1) = P \cdot \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = [\vec{p}_1 \ \dots \ \vec{p}_n] \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \vec{p}_1 = C_D(\mathbf{v}_1)$$

More generally, if $PC_B(\mathbf{v}_i) = C_D(\mathbf{v}_i)$, what must be true?

$$C_D(\mathbf{v}_i) = \vec{p}_i$$

P is called the **transition matrix/change matrix** from B to D , and is denoted $P_{D \leftarrow B}$.

Theorem: Let $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and D denote ordered bases of a vector space V . Then the change matrix $P_{D \leftarrow B}$ is given in terms of its columns by

$$P_{D \leftarrow B} = [C_D(\mathbf{v}_1) \ C_D(\mathbf{v}_2) \ \cdots \ C_D(\mathbf{v}_n)]$$

and has the property that $C_D(\mathbf{v}) = P_{D \leftarrow B} C_B(\mathbf{v})$ for all $\mathbf{v}$ in V .

Procedure to build transition matrix/change matrix

To compute the transition matrix $P_{D \leftarrow B}$ from B to D , we calculate the coordinate vector of each vector in B relative to the basis D . Then use these coordinate vectors, in order, as columns in the matrix $P_{D \leftarrow B}$.

Example: Let $B = (1, x)$ and $D = (x+1, x-1)$ be two ordered bases for P_1 , and build $P_{D \leftarrow B}$.

$$1 = \frac{1}{2}(x+1) - \frac{1}{2}(x-1) \Rightarrow C_D(1) = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$x = \frac{1}{2}(x+1) + \frac{1}{2}(x-1) \Rightarrow C_D(x) = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\Rightarrow P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Theorem: Let B and D denote ordered bases of a vector space V . Then we have the following properties:

1. $P_{B \leftarrow B} = \underline{\mathbf{I}}$

2. $P_{D \leftarrow B}$ is invertible and $P_{D \leftarrow B}^{-1} = \underline{P_{B \leftarrow D}}$

3. If E is another ordered basis of V , $P_{E \leftarrow D} P_{D \leftarrow B} = \underline{P_{E \leftarrow B}}$

Example: As in the previous example, let $B = (1, x)$ and $D = (x+1, x-1)$ be two ordered bases for P_1 . Compute $C_D(v)$ for $v = 5 - x$ both (a) directly using D and (b) using $P_{D \leftarrow B}$.

$$(a) \quad 5 - x = a_1(x+1) + a_2(x-1) = (a_1 + a_2)x + a_1 - a_2$$

$$\Rightarrow \begin{cases} a_1 + a_2 = -1 \\ a_1 - a_2 = 5 \end{cases} \Rightarrow \begin{cases} a_1 = 2 \\ a_2 = -3 \end{cases} \Rightarrow C_D(v) = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$(b) \text{ By previous ex. , } P_{D \leftarrow B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$C_D(v) = P_{D \leftarrow B} \cdot \begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Example: Let $B = (1, x)$ and $E = (2x+1, 3x-2)$. Verify that $P_{E \leftarrow B} = \begin{bmatrix} 3/7 & 2/7 \\ -2/7 & 1/7 \end{bmatrix}$, and use this to verify your answer for $C_E(x-10)$.

$$1 = \frac{3}{7} \times (2x+1) - \frac{2}{7} \times (3x-2)$$

$$x = \frac{2}{7} (2x+1) + \frac{1}{7} (3x-2)$$

$$C_E(x-10) = \begin{bmatrix} 3/7 & 2/7 \\ -2/7 & 1/7 \end{bmatrix} \cdot \begin{bmatrix} -10 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$$

Using $D = (x+1, x-1)$, use your previous answers to compute $P_{E \leftarrow D}$.

$$x+1 = a_1 \cdot (2x+1) + a_2 \cdot (3x-2)$$

$$\Rightarrow \begin{cases} 2a_1 + 3a_2 = 1 \\ a_1 - 2a_2 = 1 \end{cases} \Rightarrow \begin{cases} a_1 = \frac{5}{7} \\ a_2 = -\frac{1}{7} \end{cases}.$$

$$x-1 = b_1 \cdot (2x+1) + b_2 \cdot (3x-2)$$

$$\begin{cases} 2b_1 + 3b_2 = 1 \\ b_1 - 2b_2 = -1 \end{cases} \Rightarrow \begin{cases} b_1 = -\frac{1}{7} \\ b_2 = \frac{3}{7} \end{cases}$$

$$\text{so } P_{E \leftarrow D} = \begin{bmatrix} \frac{5}{7} & -\frac{1}{7} \\ -\frac{1}{7} & \frac{3}{7} \end{bmatrix}$$